Marwadi University, Marwadi Chandarana Group, NAAC A+	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: VSLI(01CT0514)	**Aim:** To design CMOS implementation of XOR and XNOR gate, simulate the design	
Experiment:- 5	**Date:-** 12-08-2025	**Enrollment No:-** 92301733056

CO1: Develop fundamentals in fabrication of MOSFETs and MOS transistor

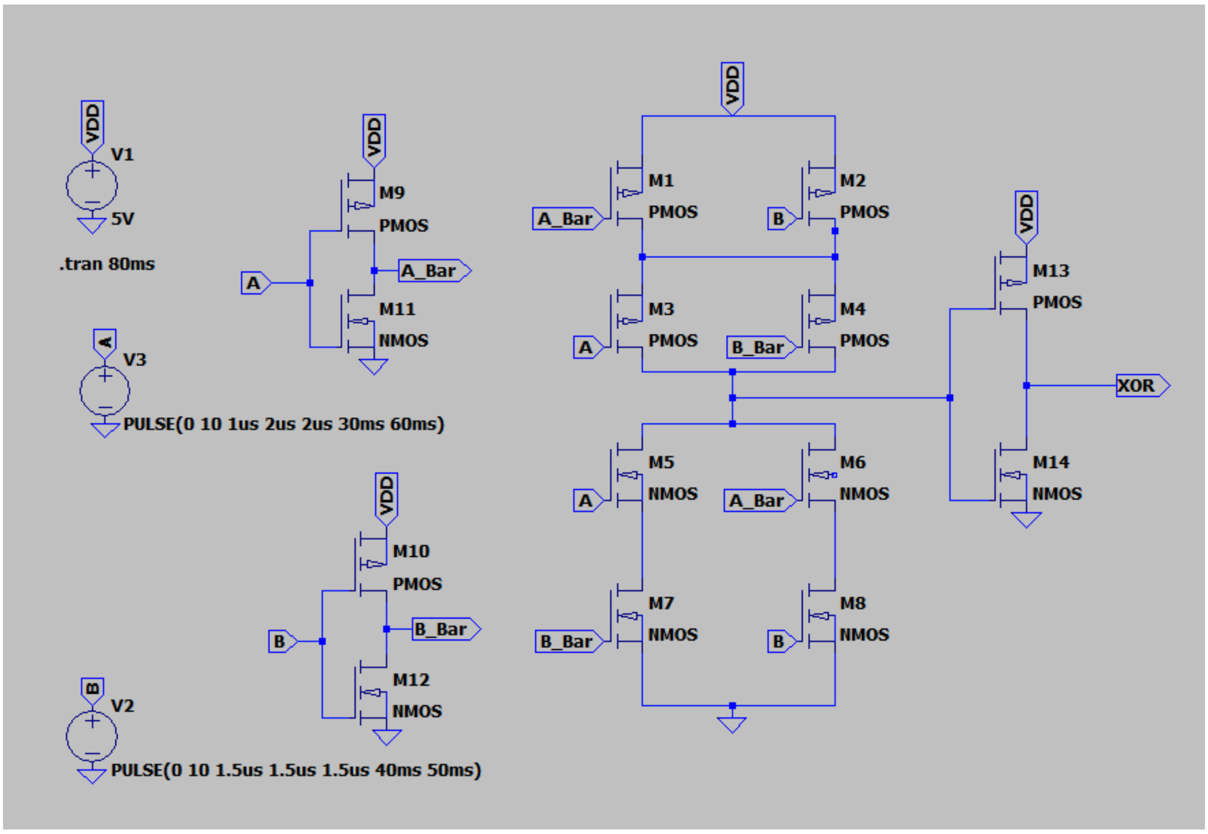
CO2: Analyze MOS inverters with its static and switching characteristics

CO3: Analyze Combinational and Sequential MOS logic circuit

AIM:-To design CMOS implementation of XOR and XNOR gate, simulate the design

Circuit:

XOR Gate:-



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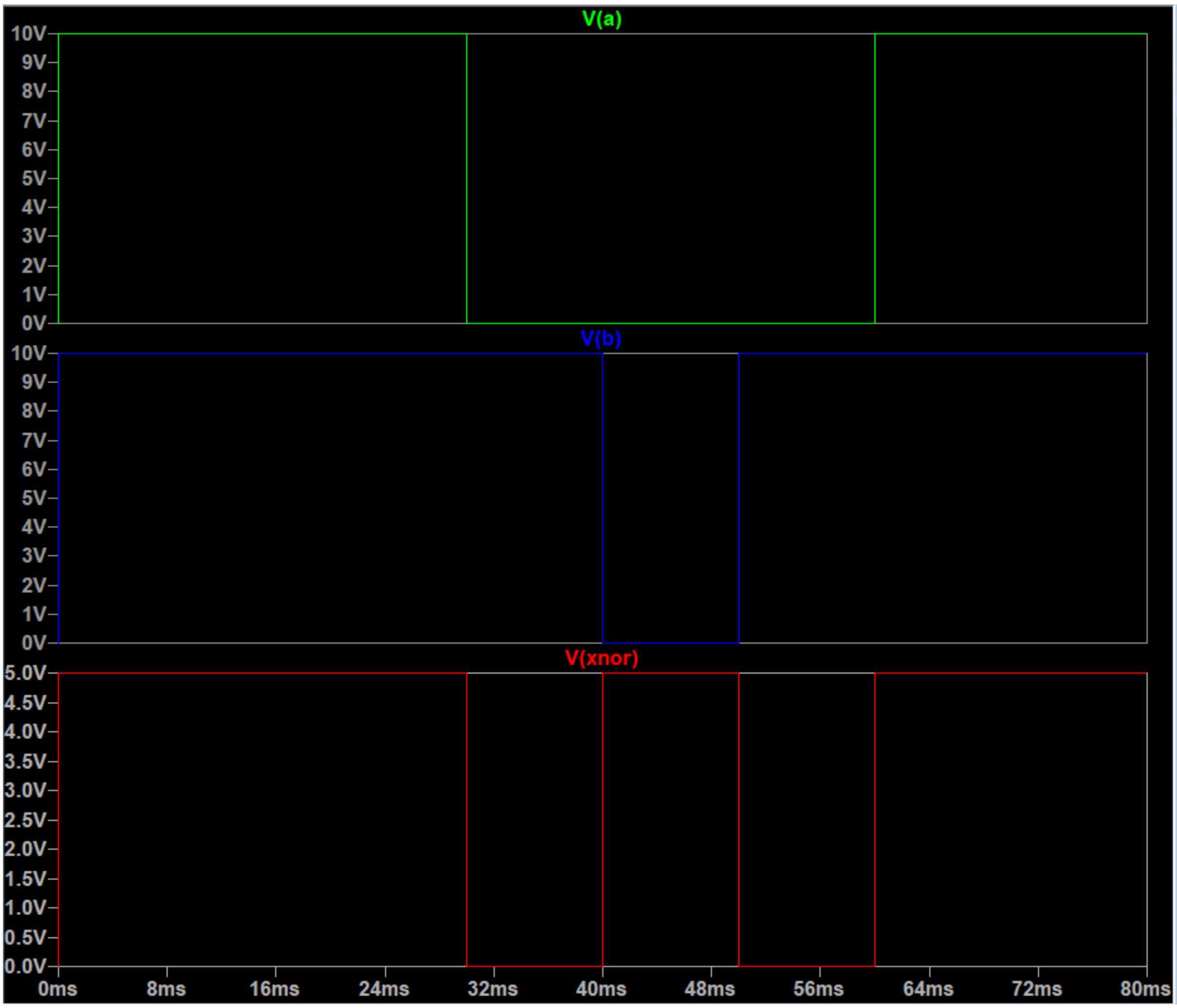
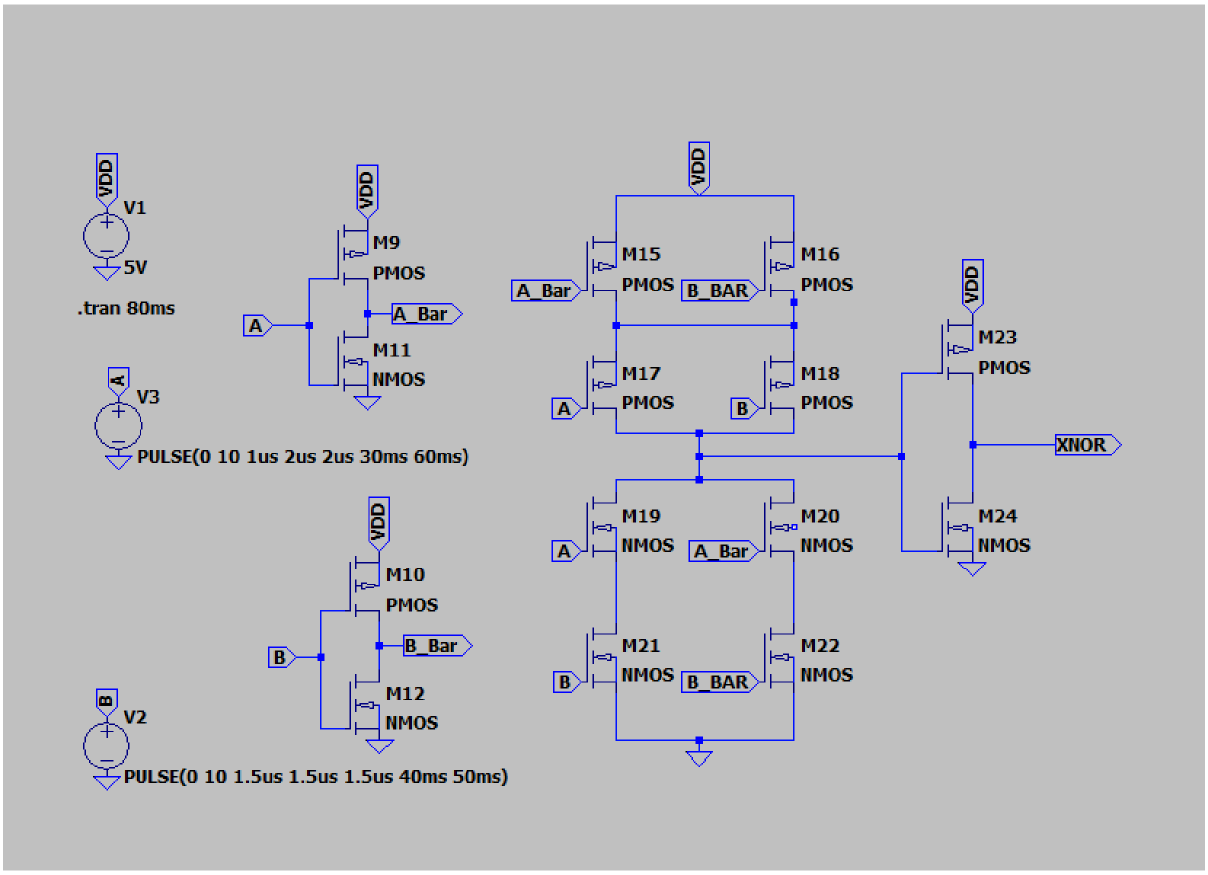
Output Waveform fo XOR

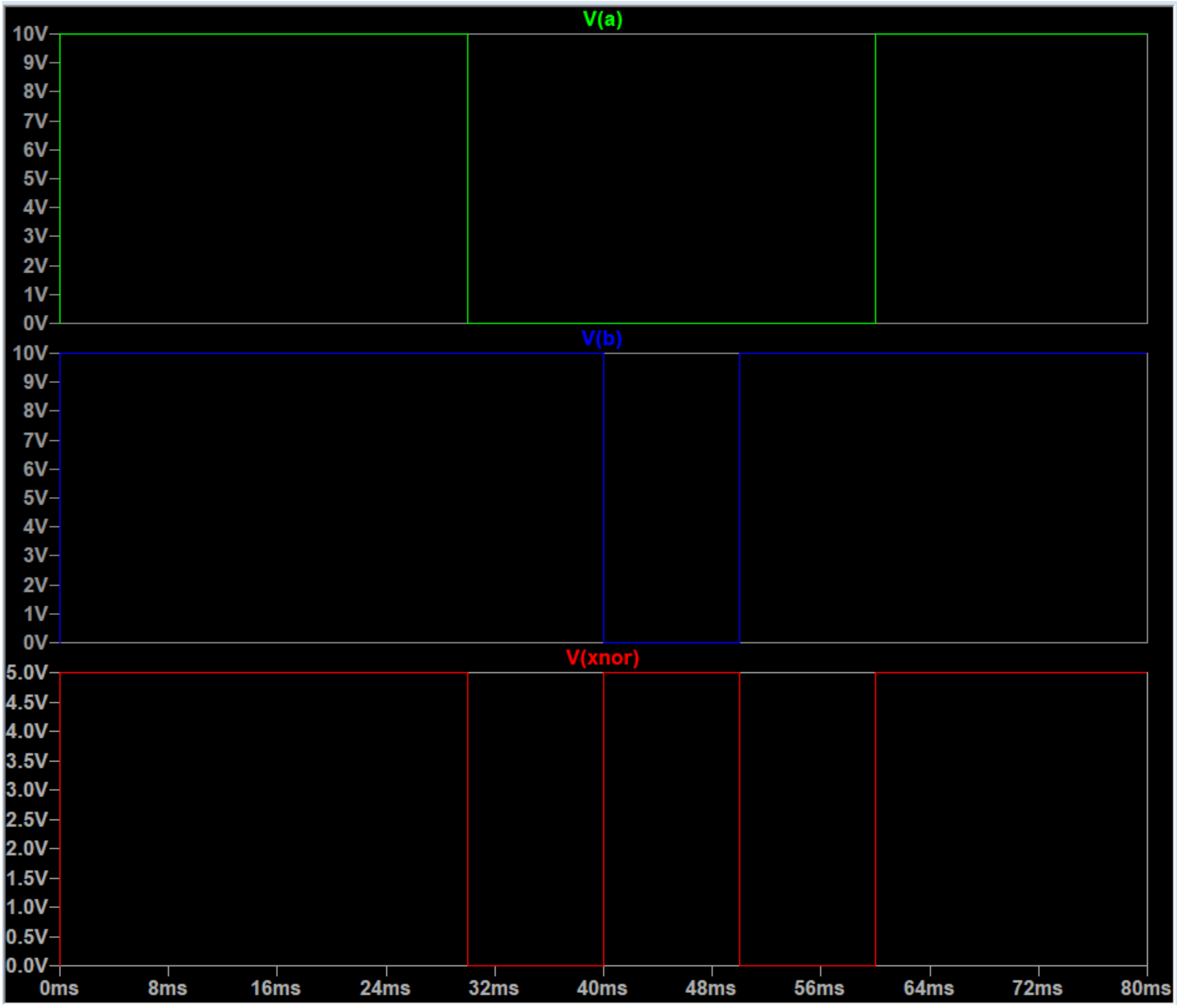


XNOR Gate:-



VDD
V1
5V
.tran 80ms
A
V3
PULSE(0 10 1us 2us 2us 30ms 60ms)
B
V2
PULSE(0 10 1.5us 1.5us 1.5us 40ms 50ms)
M9
PMOS
M11
NMOS
A_Bar
M10
PMOS
M12
NMOS
B_Bar
M15
PMOS
M16
PMOS
B_BAR
M17
PMOS
M18
PMOS
M19
NMOS
M20
NMOS
M21
NMOS
M22
NMOS
M23
PMOS
M24
NMOS
XNOR

Output Waveform for XNOR:-



Conclusion:-

We implemented the XOR and XNOR gate using CMOS(compliment-metaloxide semiconductor) and we analyzed the waveforms of the both gates..